

The L2.5 Interlayer — UPL Reversible Influence Model (RIM)

A bounded zone between L2 (execution) and L3 (representation) where forward influence (chip → output) and reverse influence (output → chip) both produce named, measurable, causally traceable operators with explicit reversibility.

FORWARD F

$L_2 \rightarrow L_3$

Forward operators (F)

Representation:
cast_precision, quantize, normalize

Cache/Memory:
cache_pin, cache_evict, prefetch

Scheduler:
select_device, split_batch, fuse_kernel

Decoding:
sample_token, rerank, beam_prune

REVERSE B

$L_3 \rightarrow L_2$

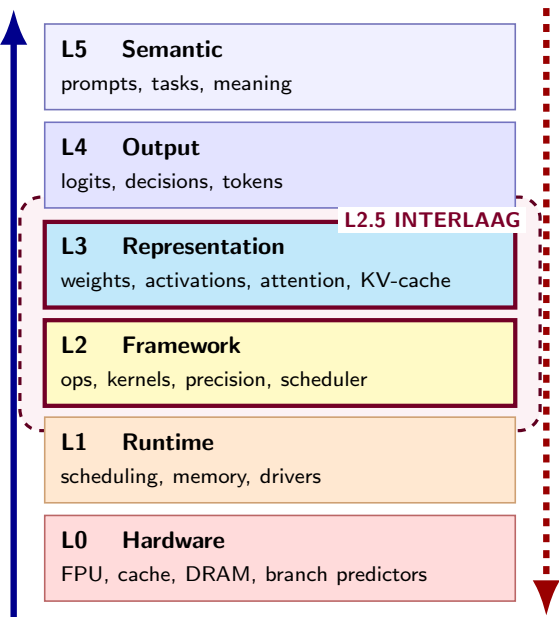
Reverse operators (B)

Replay:
replay, rollback, restore_snapshot

Compensate:
compensate, re-evaluate

Reroute:
reroute_compute, increase_precision

Stabilize:
stabilize_schedule, lock_determinism



L2.5 state $S_{L2.5}(t) = \langle C, H, M, O, R, P, N \rangle$

C compute (precision, kernel, batch, tensor layout) *H* hardware (CPU/GPU load, cache pressure, bandwidth)
M model (attention, embeddings, KV-cache, logits) *O* output (token, confidence, policy decision)
R reversibility class *P* provenance graph *N* UPL name

Forward mapping

$S_3 = F(S_2, Z)$
L2 conditions and bridge events Z
determine *L3* representational state.

Reverse mapping

$S'_2 = B(S_3, Z, \text{Policy})$
L3 state and policy modify
subsequent *L2* execution.

Reversibility classes (per event)

EXACT full snapshot restores state
SNAPSHOT checkpoint-based restore
RECONSTRUCTABLE via provenance, bounded loss
COMPENSATING correction without restore
NON_REVERSIBLE no meaningful restore

Four reversal types (per claim)

state reversal exact state restoration
causal reversal traceability of cause
operational reversal rerun with new settings
semantic reversal output re-evaluation

UPL naming convention

UPL::::<direction>::::<operator>::e.g. UPL::L2.5::FWD::PRECISION::FP32_TO_FP16::LOGIT_SHIFT
UPL::L2.5::REV::CONFIDENCE::LOW::GPU_REPLAY

Principle L2.5 Control: Every event between L2 and L3 is treated as a named, measurable, causally traceable, and possibly reversible operator. UPL adjudicates both forward and reverse claims under the same witness discipline.